# SIMATS ENGINEERING

## UNIT 3 – Solved Problems

**Course Name:** Fundamentals of Data Science

**Course Code:** DSA0404

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**Slot:** A

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**QUESTION 1**

**Find mean, median, mode and range for: 12, 15, 18, 15, 20, 22, 15, 25.**

**Solution**

Step 1: Arrange the data in ascending order

12, 15, 15, 15, 18, 20, 22, 25

Number of observations: $n = 8$

**(a) Mean**

$$\text{Mean} = \Sigma X / n$$
$$= (12 + 15 + 18 + 15 + 20 + 22 + 15 + 25) / 8$$
$$= 142 / 8$$
$$= 17.75$$

**Therefore, Mean = 17.75**

**(b) Median**

$$\text{Median} = (\text{4th value} + \text{5th value}) / 2$$
$$= (15 + 18) / 2$$
$$= 33 / 2$$
$$= 16.5$$

**Therefore, Median = 16.5**

**(c) Mode**

15 occurs 3 times, which is the highest frequency.

**Therefore, Mode = 15**

**(d) Range**

$$\text{Range} = \text{Maximum} - \text{Minimum}$$
$$= 25 - 12$$
$$= 13$$

**Therefore, Range = 13**

| Measure | Answer |
|---|---|
| Mean | 17.75 |
| Median | 16.5 |
| Mode | 15 |
| Range | 13 |

**QUESTION 2**

**Construct a frequency distribution for: 2, 3, 4, 2, 5, 3, 4, 2, 6, 5, 3, 4.**

**Solution**

Count the number of occurrences of each distinct value.

| Value (X) | Frequency (f) |
|---|---|
| 2 | 3 |
| 3 | 3 |
| 4 | 3 |
| 5 | 2 |
| 6 | 1 |
| Total | 12 |

Therefore, the frequency distribution is: 2 → 3, 3 → 3, 4 → 3, 5 → 2, 6 → 1.

**QUESTION 3**

**Find Q1, Q2, Q3 and IQR for: 5, 8, 10, 12, 15, 18, 20, 22, 25, 30.**

**Solution**

The data is already arranged in ascending order:

5, 8, 10, 12, 15, 18, 20, 22, 25, 30

There are 10 observations.

**Step 1: Find Q2 (Median)**

Q2 = (5th value + 6th value) / 2

= (15 + 18) / 2

= 16.5

**Step 2: Find Q1**

Lower half = 5, 8, 10, 12, 15
Q1 = 10

**Step 3: Find Q3**

Upper half = 18, 20, 22, 25, 30
Q3 = 22

**Step 4: Find IQR**

IQR = Q3 − Q1
= 22 − 10
= 12

**Therefore:**
**Q1 = 10, Q2 = 16.5, Q3 = 22, IQR = 12**

**QUESTION 4**

**Detect outliers using the IQR method for: 10, 12, 13, 15, 16, 17, 18, 20, 45.**
**Solution**

Data: 10, 12, 13, 15, 16, 17, 18, 20, 45
Q1 = (12 + 13) / 2 = 12.5
Q3 = (18 + 20) / 2 = 19
IQR = Q3 − Q1 = 19 − 12.5 = 6.5
Lower Fence = Q1 − 1.5(IQR)
= 12.5 − 1.5(6.5)
= 12.5 − 9.75
= 2.75
Upper Fence = Q3 + 1.5(IQR)

= 19 + 1.5(6.5)

= 19 + 9.75

= 28.75

Any value below 2.75 or above 28.75 is an outlier.

**Since 45 > 28.75, 45 is an outlier.**

| **Quantity** | **Value** |
|---|---|
| Q1 | 12.5 |
| Q3 | 19 |
| IQR | 6.5 |
| Lower Fence | 2.75 |
| Upper Fence | 28.75 |
| Outlier | 45 |

**QUESTION 5**

**Determine the skewness (asymmetry) of: 5, 6, 7, 7, 8, 8, 9, 20.**

**Solution**

Mean = (5 + 6 + 7 + 7 + 8 + 8 + 9 + 20) / 8

= 70 / 8

= 8.75

Median = (7 + 8) / 2

= 7.5

Since Mean > Median, the distribution is positively skewed (right-skewed).

**The value 20 creates a long tail toward the right.**

**QUESTION 6**

**A uniformly distributed variable ranges from 10 to 50. Find P(20 < X < 35).**

**Solution**

For a uniform distribution:

P(c < X < d) = (d − c) / (b − a)

Given: a = 10, b = 50, c = 20, d = 35

$P(20 < X < 35) = (35 - 20) / (50 - 10)$

$= 15 / 40$

$= 0.375$

**Therefore, $P(20 < X < 35) = 0.375 = 37.5\%$.**

**QUESTION 7**

**A normal distribution has mean 100 and standard deviation 15. Find the Z-score for X = 130.**

**Solution**

Given: $\mu = 100$, $\sigma = 15$, $X = 130$

$Z = (X - \mu) / \sigma$

$= (130 - 100) / 15$

$= 30 / 15$

$= 2$

**Therefore, Z = 2.**

The value 130 is 2 standard deviations above the mean.

**QUESTION 8**

**For the same normal distribution, find $P(X < 130)$.**

**Solution**

From Question 7, Z = 2.

Using the standard normal distribution table:

$P(Z < 2) = 0.9772$

**Therefore, $P(X < 130) = 0.9772 = 97.72\%$.**

**QUESTION 9**

**Given a dataset with one extreme value, compare the mean and median before and after outlier treatment.**

**Solution**

Consider the dataset:

10, 12, 14, 15, 16, 18, 20, 100

Here, 100 is an extreme value.

**Before Outlier Treatment**

Mean = (10 + 12 + 14 + 15 + 16 + 18 + 20 + 100) / 8

= 205 / 8

= 25.625

Median = (15 + 16) / 2 = 15.5

**After Removing the Outlier**

Dataset = 10, 12, 14, 15, 16, 18, 20

Mean = 105 / 7 = 15

Median = 15

| Measure | Before Treatment | After Treatment |
|---|---|---|
| Mean | 25.625 | 15 |
| Median | 15.5 | 15 |

Conclusion: The extreme value 100 significantly increases the mean, while the median changes only slightly.

**Therefore, the median is more resistant to outliers than the mean.**

**QUESTION 10**

**Perform a complete EDA on a small student-mark dataset and report central tendency, dispersion, distribution, outliers and skewness.**

**Given Dataset**

45, 52, 56, 60, 62, 65, 68, 70, 72, 75, 78, 95

Number of observations: n = 12

**A. Central Tendency**

Mean = (45 + 52 + 56 + 60 + 62 + 65 + 68 + 70 + 72 + 75 + 78 + 95) / 12

= 798 / 12

= 66.5

Median = (6th + 7th values) / 2

= (65 + 68) / 2

= 66.5

Mode: No mode, because every mark occurs only once.

**B. Dispersion**

Minimum = 45

Maximum = 95

Range = 95 − 45 = 50

Lower half = 45, 52, 56, 60, 62, 65

Q1 = (56 + 60) / 2 = 58

Upper half = 68, 70, 72, 75, 78, 95

Q3 = (72 + 75) / 2 = 73.5

IQR = Q3 − Q1 = 73.5 − 58 = 15.5

**C. Outlier Detection**

Lower Fence = Q1 − 1.5(IQR)

= 58 − 1.5(15.5)

= 58 − 23.25

= 34.75

Upper Fence = Q3 + 1.5(IQR)

= 73.5 + 1.5(15.5)

= 73.5 + 23.25

= 96.75

All values lie between 34.75 and 96.75.

**Therefore, there are no outliers.**

### D. Distribution and Skewness

Mean = 66.5 and Median = 66.5.

Since Mean ≈ Median, the distribution is approximately symmetric.

The high value 95 produces a slight tendency toward the right.

Distribution = Approximately symmetric

**Skewness = Slightly positive**

### E. Complete EDA Summary

| EDA Measure | Result |
|---|---|
| Number of students | 12 |
| Mean | 66.5 |
| Median | 66.5 |
| Mode | No mode |
| Minimum | 45 |
| Maximum | 95 |
| Range | 50 |
| Q1 | 58 |
| Q2 | 66.5 |
| Q3 | 73.5 |
| IQR | 15.5 |
| Lower Fence | 34.75 |
| Upper Fence | 96.75 |
| Outliers | None |
| Distribution | Approximately symmetric |
| Skewness | Slightly positive |

### Final EDA Conclusion

The student-mark dataset has a mean and median of 66.5 marks, indicating a balanced central tendency. The range is 50 marks, while the IQR is 15.5 marks, representing the spread of the middle 50% of the observations. The IQR method detects no outliers. Since the mean and median are equal, the distribution is approximately symmetric, with a slight tendency toward positive skewness because of the high score of 95.